Employee Performance & Promotion Analysis

Statistics, Probability & Linear Algebra Project

Dataset: employee_performance.csv, 4,000 records, 8 fields (Employee_ID, Department, Age, Salary, Projects_Completed, Working_Hours, Performance_Score, Promotion_Status).

Part A: Theory

1. Mean, Median, Mode: Mean is the sum divided by the count; median is the middle value of the sorted data; mode is the most frequent value. For salaries 30k, 40k, 40k, 50k, 90k: mean = 50k, median = 40k, mode = 40k. The 90k salary pulls the mean up but not the median, so the median describes a typical salary better when pay is skewed.

2. Range vs Variance: Range is max minus min (30k to 90k gives 60k); it uses only two values and is sensitive to outliers. Variance is the average squared deviation from the mean and uses every value, so it reflects the overall spread. Two teams can have the same range but different variances.

3. Normal vs Poisson: Normal is continuous, symmetric and bell-shaped, defined by mean and standard deviation (example: performance scores). Poisson is discrete, models counts of events in a fixed interval with rate lambda, where mean = variance (example: complaints received per week).

4. Skewness: Skewness measures asymmetry. Positive skew means a long right tail (mean > median). Workplace example: company salaries, where a few executives earn far more than most staff.

5. Conditional Probability: P(A|B) = P(A and B) / P(B), the probability of A given B has occurred. For promotions, P(Promotion | Performance_Score > 80) is the promotion chance among high performers; if it exceeds the overall rate, performance influences promotion.

6. Independent vs Mutually Exclusive: Independent events do not affect each other: P(A and B) = P(A)P(B) (birth month and promotion). Mutually exclusive events cannot occur together: P(A and B) = 0 (promoted and not promoted in the same cycle). Events with non-zero probability cannot be both.

7. Bayes' Theorem: P(A|B) = P(B|A)P(A)/P(B). It updates a probability when new evidence arrives, for example revising the chance that an employee belongs to the promoted group after seeing a high performance score. It supports decisions in screening, risk assessment and diagnosis.

8. PCA: Principal Component Analysis turns many correlated variables into a few uncorrelated principal components, ordered by how much variation they capture. It simplifies data, helps 2D visualization and reduces noise. Variables should be standardized first because Salary and Hours are on different scales.

Part B: Practical Analysis

Step 1: Central Tendency & Dispersion

Measure	Value
Salary mean	81,632.73
Salary median	80,500.00
Salary mode	74,700, 79,500
Projects_Completed variance (sample)	8.2606
Projects_Completed std deviation	2.8741

The mean salary is above the median, indicating a right-pulled distribution. Variance of projects (8.26) is close to its mean (8.04), which is typical of count (Poisson-like) data.

Step 2: Probability & Events

Probability of promotion: P(Promotion) = 570 / 4000 = 0.1425.

Department	No	Yes	Total
Finance	540	69	609
HR	357	49	406
IT	1017	183	1200
Marketing	415	78	493
Operations	444	77	521
Sales	657	114	771
Total	3430	570	4000

Conditional probability: P(Promotion | Performance_Score > 80) = 0.5310, compared with the overall 0.1425. High performers are 3.73 times more likely to be promoted, so promotion and performance are dependent. A chi-square test of Department vs Promotion gives chi2 = 8.11, p = 0.1503, so department has no significant effect at the 5% level.

Step 3: Distributions & Visualization

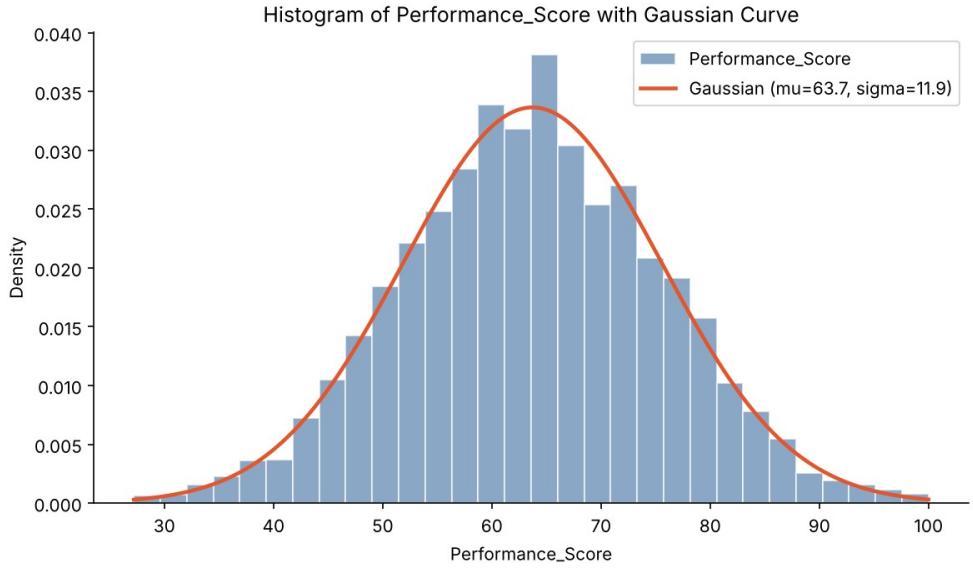


Figure 1: Performance_Score histogram with fitted Gaussian curve

Performance scores are close to normal (skewness 0.017).

Salary skewness = 1.1567 and excess kurtosis = 6.2153. Positive skewness shows a long right tail of high earners; positive kurtosis shows heavier tails than a normal distribution.

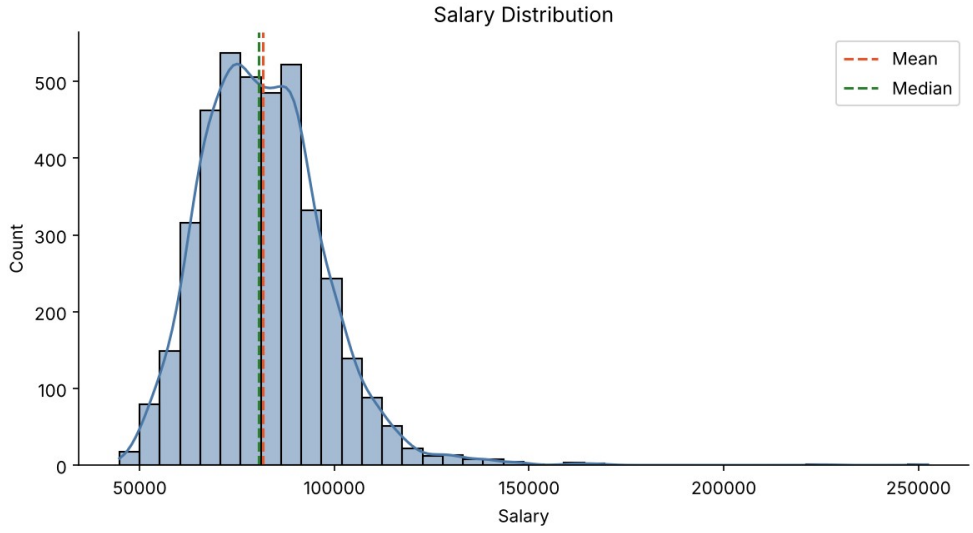


Figure 2: Salary distribution with mean and median

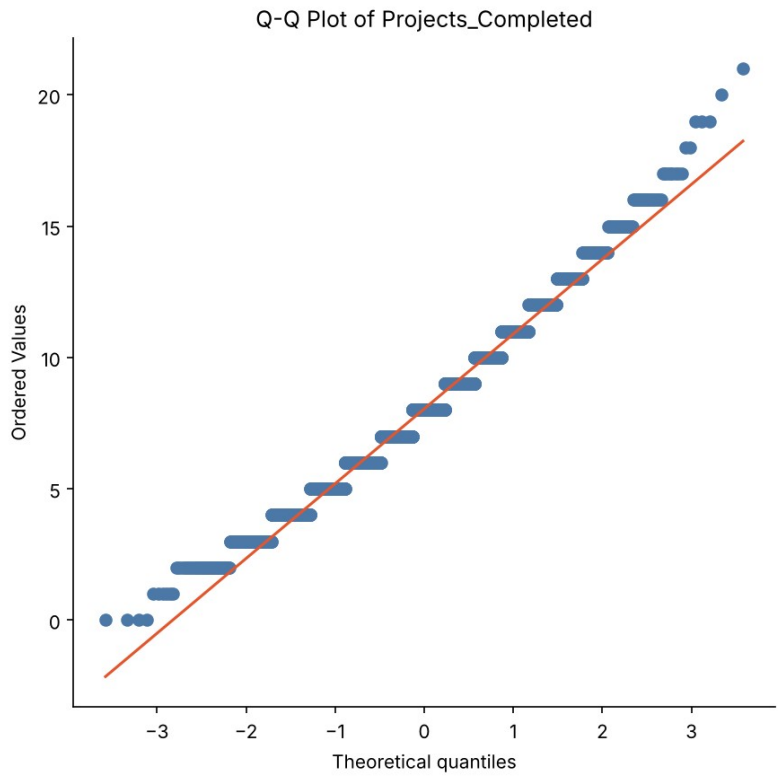


Figure 3: Q-Q plot of Projects_Completed

Points follow the reference line closely with a step pattern, because Projects_Completed is a discrete count; it is approximately normal with small tail deviations.

Step 4: Linear Algebra Application

Employee	Projects_Completed	Working_Hours
Emp 1	5	49.5
Emp 2	11	39.2
Emp 3	7	35.3
Emp 4	9	53.8
Emp 5	6	39.9

Quantity	Value
Dot product (Emp 1 . Emp 2)	1995.40
Norm 1 (L1) of Emp 1	54.5000
Norm 2 (L2) of Emp 1	49.7519
cos(theta)	0.9851
Angle between Emp 1 and Emp 2	9.91 degrees

An angle of 9.9 degrees means the two employees have a very similar balance of projects to hours, even though their magnitudes differ.

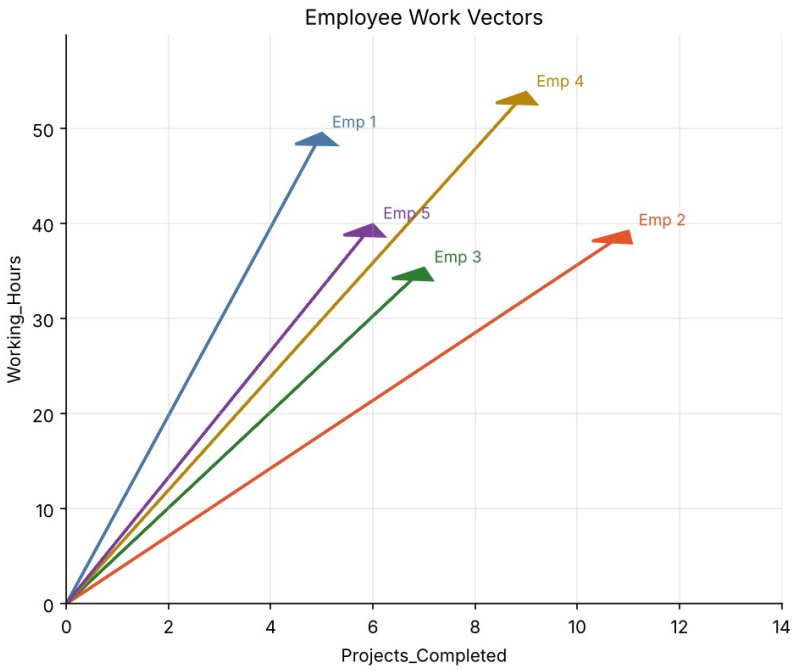


Figure 4: Work vectors of the first five employees

Additional Insights

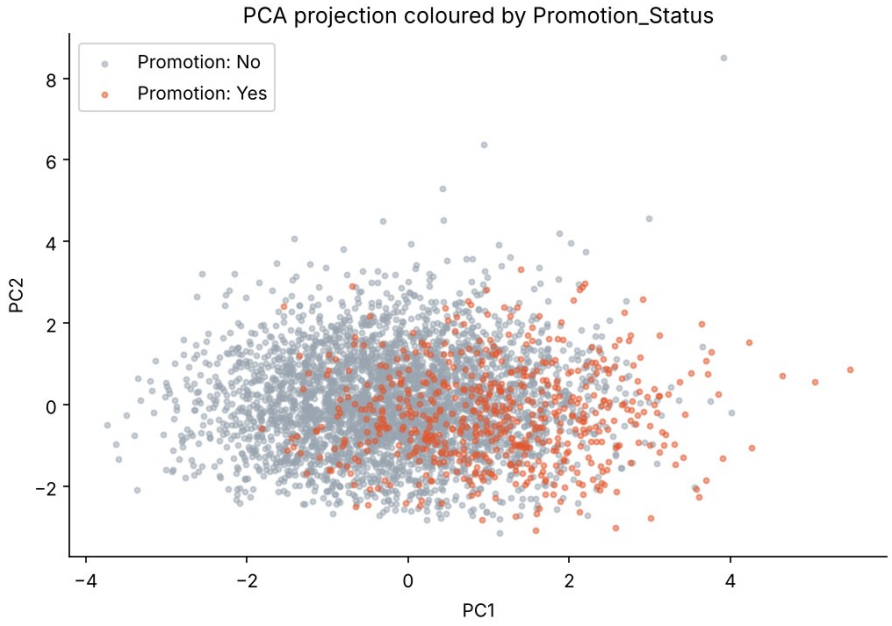


Figure 5: PCA projection (standardized numeric variables) coloured by promotion

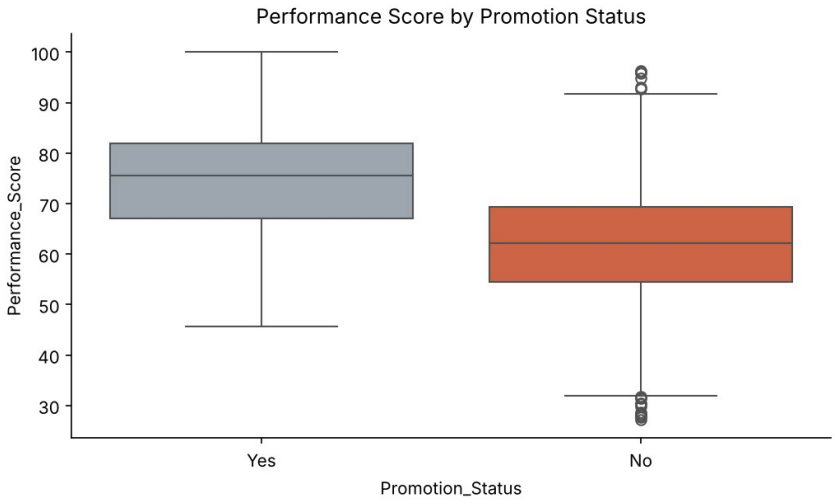


Figure 6: Performance score by promotion status

Conclusions

- Salary is right-skewed with heavy tails; the median (80,500) is a more robust centre than the mean (81,633).
- Projects_Completed behaves like count data and is only approximately normal.
- Promotion is uncommon (14.2%) but jumps to 53.1% for employees scoring above 80, so performance drives promotion.
- Vector analysis (dot product, norms, angle) quantifies how similar employees' work patterns are.
- PCA summarizes the numeric variables in fewer dimensions for visualization.

Note: figures come from the dataset file supplied to the notebook; re-run the notebook with the original data to update them.